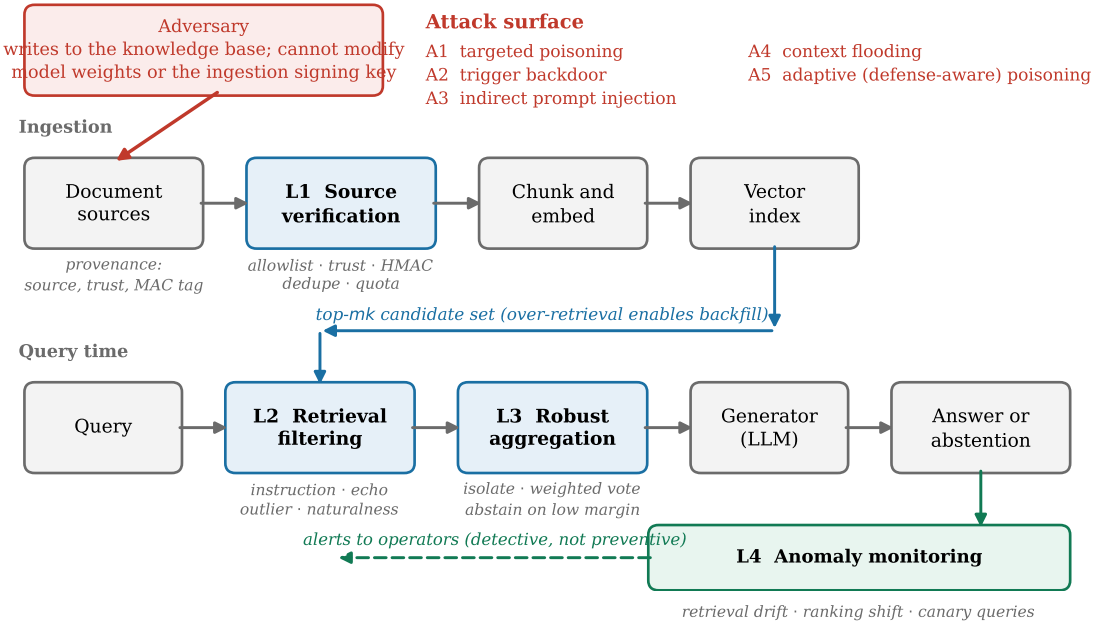
Adversary
writes to the knowledge base; cannot modify
model weights or the ingestion signing key
Attack surface
A1 targeted poisoning
A2 trigger backdoor
A3 indirect prompt injection
A4 context flooding
A5 adaptive (defense-aware) poisoning
Ingestion
Document sources
L1 Source verification
Chunk and embed
Vector index
provenance:
source, trust, MAC tag
allowlist · trust · HMAC
dedupe · quota
top-mk candidate set (over-retrieval enables backfill)
Query time
Query
L2 Retrieval filtering
L3 Robust aggregation
Generator (LLM)
Answer or abstention
instruction · echo
outlier · naturalness
isolate · weighted vote
abstain on low margin
alerts to operators (detective, not preventive)
L4 Anomaly monitoring
retrieval drift · ranking shift · canary queries